

- GitHub repository or project files

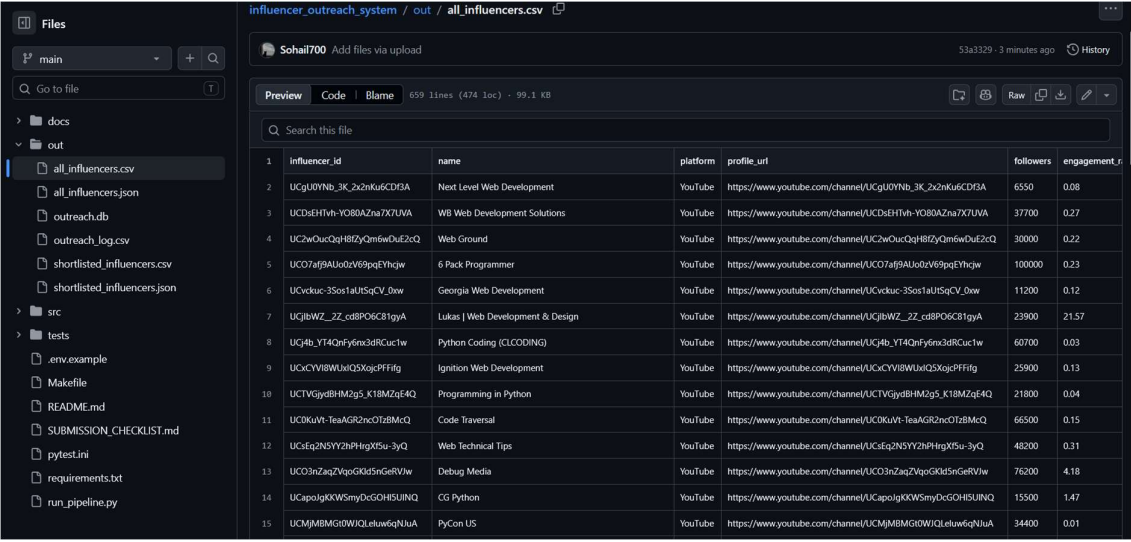
URL: https://github.com/Sohail700/influencer_outreach_system

- Working demo or screenshots/video

```
> ls
data  Makefile  pytest.ini  requirements.txt  src  tests
docs  out      README.md  run_pipeline.py  SUBMISSION_CHECKLIST.md
> python run_pipeline.py --target 100 --skip-llm
Discovered/enriched: 66
Passed filter: 22
Personalized messages: 0
Outputs written to ./out

~/influencer_outreach_system 2m 4s influencer_outreach_system 02:18:52 PM
> █
```

• Influencer dataset



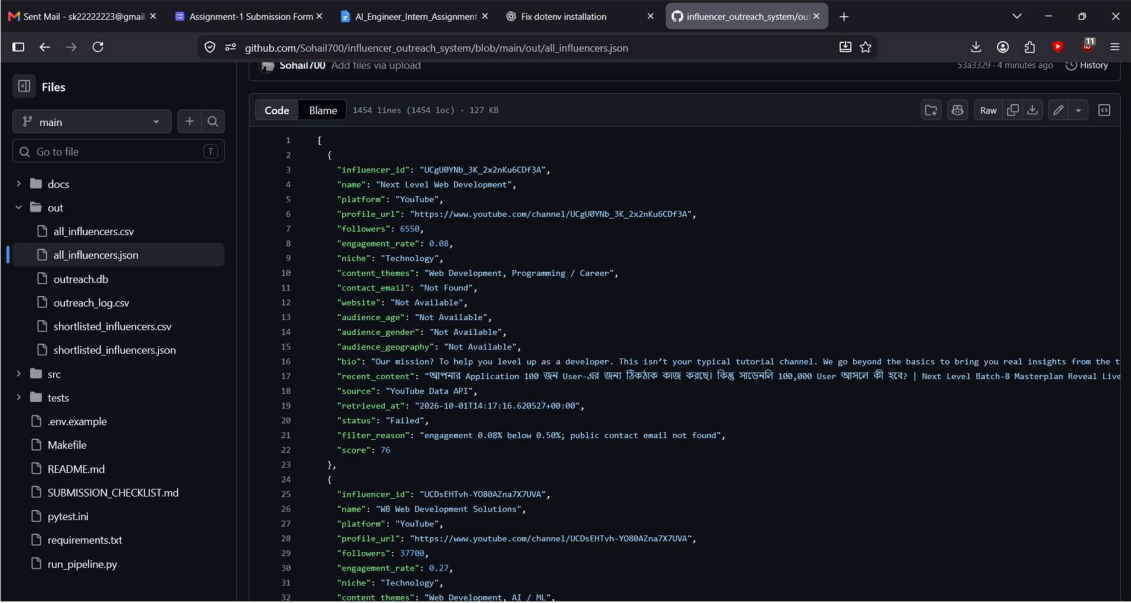
influencer_outreach_system / out / all_influencers.csv

Sohail700 Add files via upload 53a3329 - 3 minutes ago History

Preview Code Blame 659 lines (474 loc) · 99.1 KB

Search this file

	influencer_id	name	platform	profile_url	followers	engagement_r
1	UCgU0YNb_3K_2x2nKu6CD3A	Next Level Web Development	YouTube	https://www.youtube.com/channel/UCgU0YNb_3K_2x2nKu6CD3A	6550	0.08
2	UCDsEHTvh-YO80AZna7X7UVA	WB Web Development Solutions	YouTube	https://www.youtube.com/channel/UCDsEHTvh-YO80AZna7X7UVA	37700	0.27
3	UC2wOucQqH8IzOm6wOue2zCQ	Web Ground	YouTube	https://www.youtube.com/channel/UC2wOucQqH8IzOm6wOue2zCQ	30000	0.22
4	UCO7af9AUoQzV69pgEYhcjw	6 Pack Programmer	YouTube	https://www.youtube.com/channel/UCO7af9AUoQzV69pgEYhcjw	100000	0.23
5	UCvkcuc-3Sosa1uISqCV_0w	Georgia Web Development	YouTube	https://www.youtube.com/channel/UCvkcuc-3Sosa1uISqCV_0w	11200	0.12
6	UCjBwZ_2Z_cdBPOGCB1yA	Lukas Web Development & Design	YouTube	https://www.youtube.com/channel/UCjBwZ_2Z_cdBPOGCB1yA	23900	21.57
7	UC4b_YT4QnFy6n3dRCuc1w	Python Coding (CLCODING)	YouTube	https://www.youtube.com/channel/UC4b_YT4QnFy6n3dRCuc1w	60700	0.03
8	UCxCIv8WUwQXkjPFFifg	Ignition Web Development	YouTube	https://www.youtube.com/channel/UCxCIv8WUwQXkjPFFifg	25900	0.13
9	UCTVGjyBHM2gS_K18M2q4EQ	Programming in Python	YouTube	https://www.youtube.com/channel/UCTVGjyBHM2gS_K18M2q4EQ	21800	0.04
10	UC0KuVt-TeaGR2ncOTz8McQ	Code Traversal	YouTube	https://www.youtube.com/channel/UC0KuVt-TeaGR2ncOTz8McQ	66500	0.15
11	UCfEqN5Y2hPHrgX5u-3yQ	Web Technical Tips	YouTube	https://www.youtube.com/channel/UCfEqN5Y2hPHrgX5u-3yQ	48200	0.31
12	UCO3nZaqZVqoGkdsnGerVjw	Debug Media	YouTube	https://www.youtube.com/channel/UCO3nZaqZVqoGkdsnGerVjw	76200	4.18
13	UCapolgKKWsmYdCGOH5UINQ	CG Python	YouTube	https://www.youtube.com/channel/UCapolgKKWsmYdCGOH5UINQ	15500	1.47
14	UCMjMBMGdWlQleluwqNJuA	PyCon US	YouTube	https://www.youtube.com/channel/UCMjMBMGdWlQleluwqNJuA	34400	0.01



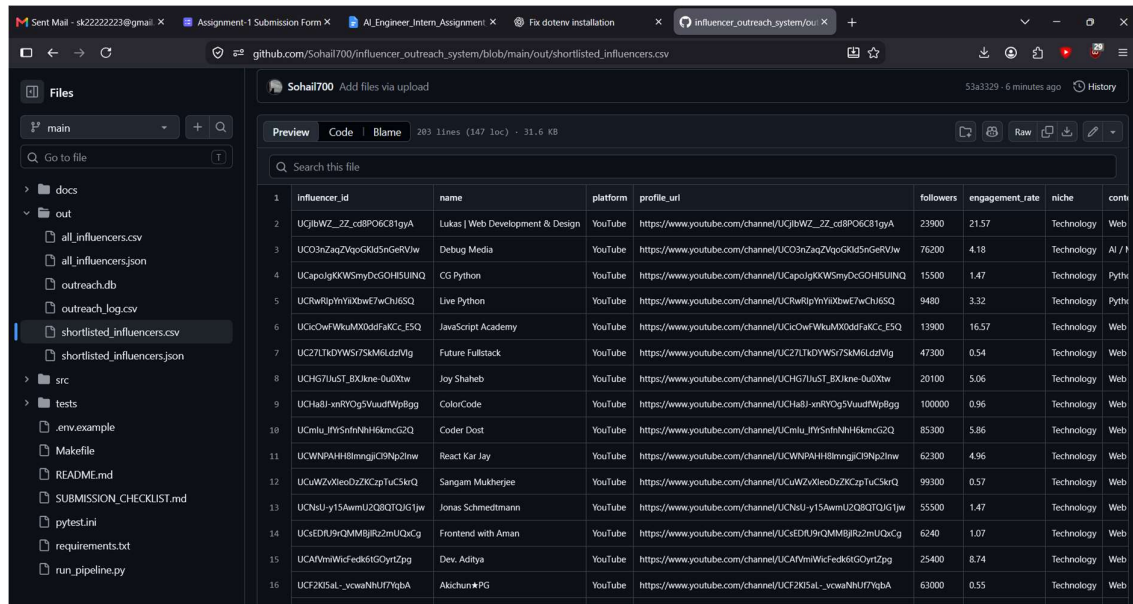
influencer_outreach_system/out/all_influencers.json

Sohail700 Add files via upload 53a3329 - 4 minutes ago History

Code Blame 1454 lines (1454 loc) · 127 KB

```
1 {
2   {
3     "influencer_id": "UCgU0YNb_3K_2x2nKu6CD3A",
4     "name": "Next Level Web Development",
5     "platform": "YouTube",
6     "profile_url": "https://www.youtube.com/channel/UCgU0YNb_3K_2x2nKu6CD3A",
7     "followers": 6550,
8     "engagement_rate": 0.08,
9     "niche": "Technology",
10    "content_themes": "Web Development, Programming / Career",
11    "contact_email": "Not Found",
12    "website": "Not Available",
13    "audience_age": "Not Available",
14    "audience_gender": "Not Available",
15    "audience_geography": "Not Available",
16    "bio": "Our mission? To help you level up as a developer. This isn't your typical tutorial channel. We go beyond the basics to bring you real insights from the t",
17    "recent_content": "আমাদের Application 100 জন User-এর জন্য বিকল্প কাজ করছে। কিন্তু মাসে মাসে 100,000 User আসলে কী হবে? | Next Level Batch-8 Masterplan Reveal Live",
18    "source": "YouTube Data API",
19    "retrieved_at": "2026-10-01T14:17:16.620527+00:00",
20    "status": "Failed",
21    "filter_reason": "engagement 0.08% below 0.50%; public contact email not found",
22    "score": 76
23  },
24  {
25    "influencer_id": "UCDsEHTvh-YO80AZna7X7UVA",
26    "name": "WB Web Development Solutions",
27    "platform": "YouTube",
28    "profile_url": "https://www.youtube.com/channel/UCDsEHTvh-YO80AZna7X7UVA",
29    "followers": 37700,
30    "engagement_rate": 0.27,
31    "niche": "Technology",
32    "content_themes": "Web Development, AI / ML",
```

- Sample personalized outreach messages



The screenshot shows a GitHub repository for 'Sohail700' with a file named 'shortlisted_influencers.csv'. The file is a CSV with 16 rows of data. The columns are: Influencer_id, name, platform, profile_url, followers, engagement_rate, niche, and content. The data lists various YouTube influencers in the technology and web development space.

Influencer_id	name	platform	profile_url	followers	engagement_rate	niche	content
UCjBwZ_2Z_cdBPO6C81gyA	Lukas Web Development & Design	YouTube	https://www.youtube.com/channel/UCjBwZ_2Z_cdBPO6C81gyA	23900	21.57	Technology	Web
UCO3nZaqZVqoGKd5nGeRVjw	Debug Media	YouTube	https://www.youtube.com/channel/UCO3nZaqZVqoGKd5nGeRVjw	76200	4.18	Technology	AI / ML
UCapogRWsmyDxGOHSUINQ	CG Python	YouTube	https://www.youtube.com/channel/UCapogRWsmyDxGOHSUINQ	15500	1.47	Technology	Python
UCRwRlpYnYixbwE7wCh6SQ	Live Python	YouTube	https://www.youtube.com/channel/UCRwRlpYnYixbwE7wCh6SQ	9480	3.32	Technology	Python
UCicOwWkuMX0ddFaKc_E5Q	JavaScript Academy	YouTube	https://www.youtube.com/channel/UCicOwWkuMX0ddFaKc_E5Q	13900	16.57	Technology	Web
UC27LTKDWS73kM6LdtVlg	Future Fullstack	YouTube	https://www.youtube.com/channel/UC27LTKDWS73kM6LdtVlg	47300	0.54	Technology	Web
UCHG7JusT_BXIkne-0u0Xtw	Joy Shaheb	YouTube	https://www.youtube.com/channel/UCHG7JusT_BXIkne-0u0Xtw	20100	5.06	Technology	Web
UCHa8I-xmRYOgSVuudfWpBgg	ColorCode	YouTube	https://www.youtube.com/channel/UCHa8I-xmRYOgSVuudfWpBgg	100000	0.96	Technology	Web
UCmlu_JW5nfnNH6kmcG2Q	Coder Dost	YouTube	https://www.youtube.com/channel/UCmlu_JW5nfnNH6kmcG2Q	85300	5.86	Technology	Web
UCWNP4H8ImngjC9Np2lnw	React Kar Jay	YouTube	https://www.youtube.com/channel/UCWNP4H8ImngjC9Np2lnw	62300	4.96	Technology	Web
UCuWZvXleOdzZXCzpTucSkRQ	Sangam Mukherjee	YouTube	https://www.youtube.com/channel/UCuWZvXleOdzZXCzpTucSkRQ	99300	0.57	Technology	Web
UCN6U-y15AwmlU2QBQ7QIG1jw	Jonas Schmedtmann	YouTube	https://www.youtube.com/channel/UCN6U-y15AwmlU2QBQ7QIG1jw	55500	1.47	Technology	Web
UCsEDU9rQMMBjRz2mUQcG	Frontend with Aman	YouTube	https://www.youtube.com/channel/UCsEDU9rQMMBjRz2mUQcG	6240	1.07	Technology	Web
UCANmWicFed86GOyr1Zpg	Dev. Aditya	YouTube	https://www.youtube.com/channel/UCANmWicFed86GOyr1Zpg	25400	8.74	Technology	Web
UCF2K5aL_ycwaNHU77qba	Akichun★PG	YouTube	https://www.youtube.com/channel/UCF2K5aL_ycwaNHU77qba	63000	0.55	Technology	Web

- Setup instructions
- List of APIs/tools used

Setup Instructions

1. Prerequisites

Before running the AI Micro-Influencer Outreach System, the following software and credentials are required:

- Python 3.11 or newer
- Git
- Internet connection
- YouTube Data API v3 key
- OpenAI API key for AI-powered personalization
- SMTP credentials for actual email sending (optional)

The project was developed and tested using Python and can be run on Windows, Linux, or WSL.

2. Clone or Open the Project

Clone the project repository from GitHub:

```
git clone <YOUR_GITHUB_REPOSITORY_URL>
```

```
cd influencer_outreach_system
```

If the project is already available locally, simply open a terminal inside the project directory.

3. Create a Virtual Environment

A Python virtual environment is used to keep the project's dependencies isolated from the system Python installation.

Create the virtual environment:

```
python3 -m venv .venv
```

Activate it on Linux, WSL, or macOS:

```
source .venv/bin/activate
```

On Windows PowerShell:

```
.venv\Scripts\Activate.ps1
```

4. Install Dependencies

After activating the virtual environment, install the required Python packages:

```
pip install -r requirements.txt
```

The required packages include libraries for API communication, data processing, environment-variable management, the Streamlit dashboard, and automated testing.

5. Configure Environment Variables

The project uses environment variables to store API credentials and configuration values.

Create a .env file in the project root. A .env.example file can be used as a template if provided.

Example configuration:

```
YOUTUBE_API_KEY=your_youtube_api_key
```

```
OPENAI_API_KEY=your_openai_api_key
```

```
OPENAI_MODEL=your_configured_openai_model
```

```
MIN_FOLLOWERS=5000
```

```
MAX_FOLLOWERS=100000
```

```
MIN_ENGAGEMENT_RATE=0.5
```

```
MIN_SCORE=60
```

```
DISCOVERY_TARGET=50
```

```
RECENT_VIDEO_COUNT=8
```

SMTP_HOST=smtp.gmail.com

SMTP_PORT=587

SMTP_USERNAME=your_sender@gmail.com

SMTP_PASSWORD=your_app_password

SMTP_FROM=your_sender@gmail.com

The values should be replaced with the user's own API keys and configuration.

The .env file contains sensitive information and must not be committed to GitHub.

6. Configure the YouTube API

The YouTube Data API v3 is required for influencer discovery.

General setup steps:

1. Open Google Cloud Console.
2. Create or select a Google Cloud project.
3. Open the API Library.
4. Search for **YouTube Data API v3**.
5. Enable the API.
6. Open the Credentials section.
7. Create an API key.
8. Copy the key into the .env file.

Example:

YOUTUBE_API_KEY=your_api_key

The API is used to search for technology-related channels and retrieve channel and video information.

7. Configure the OpenAI API

The OpenAI API is used to generate personalized outreach messages.

General setup steps:

1. Create or sign in to an OpenAI developer account.
2. Open the API key management section.
3. Create an API key.
4. Add the key to the .env file.

Example:

OPENAI_API_KEY=your_api_key

The configured OpenAI model is used to generate personalized email and Instagram DM content based on the influencer information collected by the pipeline.

An OpenAI API key is separate from API usage credits or billing availability. The account/project must have available API usage for the personalization stage to run successfully.

8. Configure SMTP (Optional)

SMTP is only required when actual email sending is needed.

For example, Gmail SMTP can be configured using:

SMTP_HOST=smtp.gmail.com

SMTP_PORT=587

SMTP_USERNAME=your_sender@gmail.com

SMTP_PASSWORD=your_app_password

SMTP_FROM=your_sender@gmail.com

For Gmail, an App Password should be used when required instead of the normal account password.

Actual email sending is optional. The project supports dry-run/simulation mode so the outreach workflow can be tested without sending real emails.

9. Run the Pipeline

After configuration, activate the virtual environment and run:

```
python run_pipeline.py --target 50
```

This starts the complete workflow:

Discovery → Data Collection → Filtering → Enrichment → AI Personalization → Outreach Processing → Output Generation

10. Run Without AI Personalization

For testing the discovery, enrichment, and filtering stages without calling the OpenAI API:

```
python run_pipeline.py --target 50 --skip-llm
```

This is useful during development or when the OpenAI API is unavailable.

11. Run the Streamlit Dashboard

To launch the project dashboard:

```
streamlit run app.py
```

The dashboard provides a visual way to inspect influencers, filtering results, personalized messages, and outreach tracking information.

12. Run Tests

The automated test suite can be executed using:

`pytest -q`

This verifies important components such as filtering logic and personalization validation.

13. Security Precautions

The following information must never be committed to GitHub or shared publicly:

- API keys
- SMTP passwords
- .env files
- Access tokens
- Other private credentials

The virtual environment should also not be committed.

Recommended .gitignore entries include:

`.env`

`.venv/`

`__pycache__/`

`*.pyc`

`.pytest_cache/`